Due: February 25th, before class.

Name: Solution

Collaborators:

Document your work and note any collaborators. In your write-up, please neatly document any code you have written, alongside its output, and including any necessary commentary to answer the question. Please use an `.ipynb` notebook for any code, include commentary in markdown code blocks, and print the `.ipynb` notebook to PDF to attach to your submission. The LaTeXsource for this file may be helpful: https://www.overleaf.com/read/zssmfrvwbntv#3eb27a.

Problems:

1. Let $\mathbf{P}$ be a Householder reflector defined by $\mathbf{v} \neq \mathbf{0}$.

   (a) Find a vector $\mathbf{u}$ such that $\mathbf{Pu} = -\mathbf{u}$.
   If $\mathbf{Pu} = -\mathbf{u}$, then we have
   $$\left(\mathbf{I} - 2\mathbf{v}\mathbf{v}^T\right)\mathbf{u} = -\mathbf{u},$$
   or
   $$\mathbf{u} = \mathbf{v}\mathbf{v}^T\mathbf{u}.$$
   Since $\mathbf{v}\mathbf{v}^T$ projects $\mathbf{u}$ onto its span and the result after the projection is still $\mathbf{u}$, we see that $\mathbf{u}$ must be in the span of $\mathbf{v}$, or $\mathbf{u} = k\mathbf{v}$ for any k.

   (b) What algebraic condition is necessary and sufficient for a vector $\mathbf{x}$ to satisfy $\mathbf{Px} = \mathbf{x}$? The space of all such vectors is a subspace. In n dimensions, what dimension is this subspace?
   $\mathbf{Px} = \mathbf{x}$ iff
   $$\left(\mathbf{I} - 2\mathbf{v}\mathbf{v}^T\right)\mathbf{x} = \mathbf{x},$$
   or
   $$\mathbf{v}\mathbf{v}^T\mathbf{x} = \mathbf{0}.$$
   Since $\mathbf{v} \neq \mathbf{0}$, we must have $\mathbf{v}^T\mathbf{x} = 0$. This suggests that $\mathbf{x}$ and $\mathbf{v}$ are mutually orthogonal and thus $\mathbf{x}$ lives in the hyperplane with normal vector defined by $\mathbf{v}$, which is in $n-1$ dimension.

2. Suppose you want to evaluate the inverse of the function $h(x)$; that is, given an input y, you want to evaluate $x = h^{-1}(y)$. Reformulate this problem as a root-finding problem, $f(x) = 0$, where f is defined only in terms of the function h and input y.

   If $y = h(x)$ then it means that $h(x) - y = 0$. So $f(x) = h(x) - y = 0$. If we have a root x_0 for f then it means that $y = h(x_0)$ and so $x_0 = h^{-1}(y)$.

3. The **bisection method** is a *bracketing* method for rootfinding, which keeps track of two iterates between which a root is known to exist. Consider the rootfinding problem $f(x) = 0$ defined by continous function f. Suppose that $a_1 < b_1$ are such that $f(a_1)f(b_1) < 0$. Without loss of generality, assume $f(a_1) < 0$ and $f(b_1) > 0$. The intermediate value theorem guarantees that there exists a root $r \in [a_1, b_1]$. For $k = 1, 2, \cdots$, the bisection method sets

$$x_k = \frac{a_k + b_k}{2}.$$

The interval brackets are updated as

$$\begin{cases} b_k = x_{k-1}, a_k = a_{k-1} & \text{if } f(x_{k-1}) > 0 \\ b_k = b_{k-1}, a_k = x_{k-1} & \text{otherwise.} \end{cases}$$

Suppose $[a_1, b_1]$ contains one root of f, r. Define $\epsilon_k = b_k - a_k$.

(a) Prove that $|x_k - r| \le |\epsilon_k|$ for each k.
$|x_k - r|$ represents the distance from our estimate to the actual root. $|\epsilon_k|$ represents the width of the search interval. The left hand side represents an interval within the search domain, thus by definition its size is less than or equal to the size of the search domain.

(b) Prove that the bisection enjoys at least linear convergence; $|\epsilon_{k+1}| \le \sigma|\epsilon_k|$ for some $\sigma < 1$.
We know that

$$|\epsilon_{k+1}| = |b_{k+1} - a_{k+1}| = \begin{cases} \left|\frac{a_k+b_k}{2} - a_k\right| = \left|\frac{b_k-a_k}{2}\right| & f(x_k) > 0 \\ \left|\frac{a_k+b_k}{2} - b_k\right| = \left|\frac{a_k-b_k}{2}\right| & \text{otherwise.} \end{cases}$$

Regardless of which case, we see that

$$|\epsilon_{k+1}| = \frac{1}{2}|b_k - a_k| = \frac{1}{2}|\epsilon_k|.$$

Thus

$$|\epsilon_{k+1}| \le \sigma|\epsilon_k|$$

with $\sigma \in [\frac{1}{2}, 1)$.

(c) What is σ for the bisection method?
$\sigma = \frac{1}{2}$ works.

Lab: We have seen rootfinding methods that have linear, superlinear, and quadratic convergence. We know that the bisection method has at least linear convergence, the secant method has superlinear convergence, and Newton's method has quadratic convergence. These are ordered from slowest to fastest. In fact, quadratic convergence can be dramatically faster than linear convergence. Today we'll explore this fact by implementing these methods, testing them on a given rootfinding problem, and plotting the errors!

Goals: You will visualize how linear, superlinear, and quadratic convergence lead to very different error plots and total number of iterations.

Procedure: Perform the following steps.

(a) Implement the bisection method in Julia. Make sure to return a vector `x` that contains as entries the approximations x_k.

(b) Define $f(x) = \frac{1}{x-3} - 6$ in Julia and create a plot of this function over the interval $[3.1, 3.4]$. Notice that there is a root at $19/6$.

(c) Apply the bisection method (initialized with $a_1 = 3.1$ and $b_1 = 3.5$), the secant method (initialized with $x_1 = 3.5$ and $x_2 = 3.1$), and Newton's method (initialized with $x_1 = 3.1$) to approximate this root. You may copy the code for the secant method and Newton's method from our class notes. (Each of these methods should converge with this proper initialization.)

(d) Let $r = 19/6$ and calculate the errors $\epsilon_k = x_k - r$ for each method. (You should have a vector of errors for each method, each with as many entries as iterations were run for that method.)

(e) Create a single log-linear plot (yscale is logarithmic) which plots the magnitude of the errors for each of these methods. (Exclude any error values that are 0.)